Management Performance:

Hea 16 Indoor air quality management

6 Credits

No Minimum Standard

Aim

To encourage and support healthy internal environments with good indoor air quality.

Question

Are there management processes in place to help maintain good levels of indoor air quality within the asset?

Credits	Answer	Select all answers that apply
0	A.	Question not answered
0	B.	No
1	C.	Yes, provision of information or training to asset users on how to operate and manage the asset's ventilation systems
1	D.	Yes, procedures and plans for cleaning the interior of the asset
1	E.	Yes, procedures and plans for inspecting the cleanliness of and for cleaning ventilation system components
1	F.	Yes, procurement policies and operation and maintenance procedures specify products that have low or no emissions of pollutants to air
1	G.	Yes, procedures and plans for regularly monitoring indoor air quality in occupied spaces
1	H.	Yes, procedures or plans that minimise the impacts on the asset's indoor air quality during maintenance, redecoration, refurbishment or construction activities on the asset

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Information provided for general asset users must: • Cover the correct use of ventilation system controls (including openable windows) where provided in occupied spaces, in order to maintain acceptable levels of indoor air quality. • How to report indoor air quality problems to the asset management. • Be included in the Building User Guide produced for Man 01 Building user guide or, where the credits for Man 01 are not achieved, in other written documentation provided to asset users. Information and training provided for asset management staff and relevant contractors must: • Cover the technical operation and maintenance of all ventilation systems and associated components. • Include procedures detailing the action(s) to be taken in the event of problems being identified with indoor air quality.	C
2.	Cleaning procedures and plans include the following: • Extent and frequency of cleaning, i.e. daily, weekly and monthly tasks, including dated cleaning records. • Regular deep cleaning of relevant areas, e.g. carpet, mats and flooring in regularly used areas such as entrances and exits, stairways, lifts, toilets, etc. • Provision of appropriate cleaning equipment and materials that minimise impacts to indoor air quality, e.g. HEPA vacuum cleaners, lint free cloths and dusters, cleaning chemicals and products (see Methodology), etc. • Training requirements and records for cleaning personnel covering cleaning methods and use of equipment and materials.	D
3.	Inspection and cleaning procedures and plans include the following where present: • Air Handling Units (AHUs) • Ductwork • Filters • Humidifiers • Heating and cooling coil surfaces • Heat recovery units • Air intakes, extracts and exhausts • Terminal units • Decentralised air treatment units, e.g. fan-coil units and induction units.	E

Criterion	Assessment criteria	Applicable Answer
	Inspection and cleaning frequencies should be in accordance with the following standards: • EN 15780:2011 Ventilation for buildings - Ductwork - Cleanliness of ventilation systems, or • Table 8.2 of ANSI/ASHRAE Standard 62.1-2016 Ventilation for Acceptable Indoor Air Quality. Alternatively, where local standards have similar requirements to either of the above standards, the local standard may be used to demonstrate compliance with this criterion.	
4.	Products that should be covered by policies and procedures include, but are not limited to: • Interior paints and coatings • Interior adhesives and sealants • Flooring materials including carpet • Furniture • Cleaning products. As a minimum, policies and procedures must cover emissions of very volatile organic compounds (VVOCs), including formaldehyde, and volatile organic compounds (VOCs). The policies and procedures should stipulate appropriate selection criteria for low or no emission products, e.g. reference to specific local standards, testing protocols or product labelling initiatives (see Methodology).	F
5.	Procedures and plans for monitoring indoor air quality shall incorporate and document the following measures conducted at least annually: • Measure the concentrations of relevant indoor air pollutants within the asset using robust monitoring and testing methodologies at representative sampling locations. Real time sensor networks may be used in lieu of point sampling. As a minimum, the monitoring should cover carbon dioxide and at least two other pollutants, e.g. particulate matter, total volatile organic compounds (TVOC), formaldehyde, carbon monoxide, nitrogen oxides (NOx) or radon. • Perform occupant surveys that include asset users' perceptions of indoor air quality. • Conduct inspections of the asset's envelope, plumbing, and HVAC equipment to identify sources of moisture and potential for condensation. • Evaluate the asset's ventilation rates, including air flows at intakes and exhausts.	G

Criterion	Assessment criteria	Applicable Answer
	The procedures and plans must also include: • Operation and maintenance of a system for logging and addressing indoor air quality complaints from asset users. • Action(s) to be taken in the event of the above monitoring identifying any problems.	
6.	Procedures and plans covering maintenance, redecoration, refurbishment or construction activities should include, but are not limited to, the following measures where appropriate: • Avoiding running ventilation systems where possible during works. • Closing and covering air intakes and vents and sealing ductwork prior to works commencing. • Using tools with dust guards and/or collectors fitted with appropriate HEPA filters to capture dust and particulate matter generated during works. • Regularly cleaning work areas throughout the works and increasing cleaning schedules for occupied areas. • Cleaning HVAC ductwork and changing filters during and after completion of the works. • Planning and coordinating works to minimise disruption to occupied areas. • Isolating work areas from other spaces by sealing doorways and windows or through physical separation (e.g. temporary barriers). • Maintaining occupied spaces under positive pressure relative to the outside and to internal areas undergoing works. • Implementing measures to avoid the tracking of dirt and pollutants from work areas to occupied areas (e.g. use of mats at entrances and exits, separation of access routes for occupants and works).	H
7.	All policies, procedures and plans must be reviewed at least annually or sooner in the event of significant changes to the number of asset users, changes in working practices or a change in use of space.	C – H

Methodology

Low or no emission products

There are a wide range of local and international standards, testing protocols and labelling initiatives for low emission products. The uptake of such initiatives, and therefore the availability of low or no emission products on the market, will vary between countries. Therefore, policies and procedures should reference local or international product initiatives that are active in the asset's location. Examples of such initiatives include but are not limited to:

- Ausschuss zur gesundheitlichen Bewertung von Bauprodukten (AgBB) evaluation scheme
- Green Label Plus™
- GUT Label